

# Missing Data

Project: Transparent Book Recommender

## 1. Your mission

### Your learning target

I can detect absent and unusable values.

I can keep unknown distinct from zero.

I can report accepted and rejected records.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

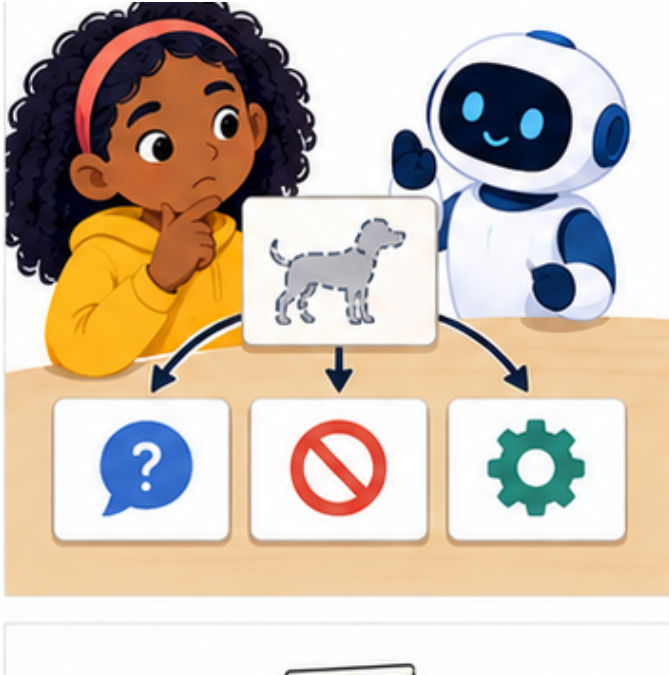
Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 18: Missing Data

A system needs an explicit policy for unknown fields.



### THINKER MISSION

Compare ask, abstain, and default strategies.

Thinker name and date:

**Write or draw your reasoning and evidence.**

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### Readiness for the connected explanation

Is the text "12" the same type as the integer 12?

### This week's end goal

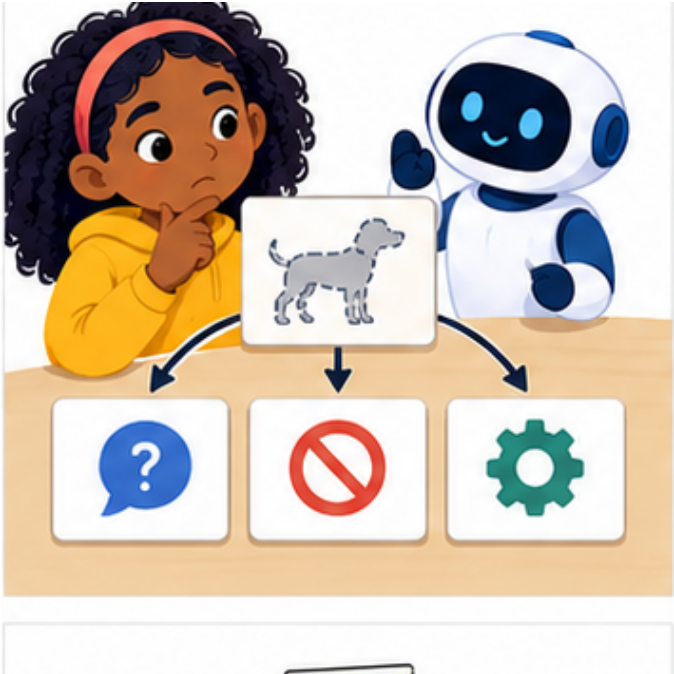
By the end, I can validate missing values without confusing zero with absent data.

## 2. Notice the evidence

### PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use missing data to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

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My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

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## A closer explanation

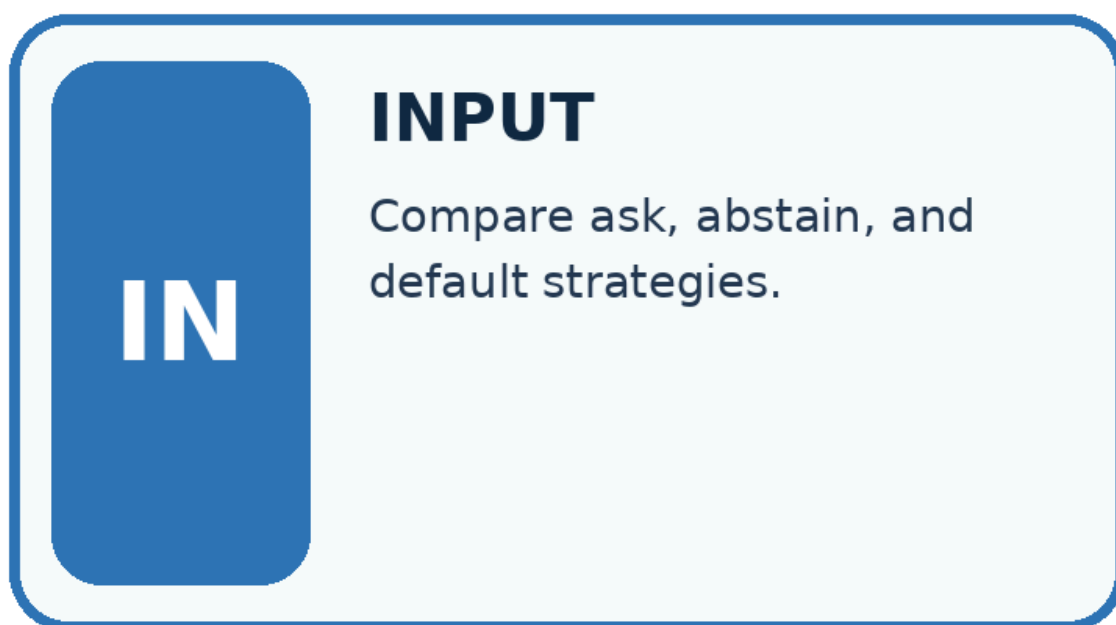
A fictional activity log contains an absent duration, a recorded zero and an out-of-range value. Our schema accepts exact integer minutes from 0 through 60. Keep rejected records visible for review instead of replacing their values with guesses.

## 3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



## PROCESS

A system needs an explicit policy for unknown fields.





## OUTPUT

Compare ask, abstain, and default strategies.

Which stage could fail even when the other two are correct? Explain:

**Write or draw your reasoning and evidence.**

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## A closer explanation

```
rows = [{"minutes": 0}, {}, {"minutes": 12},
        {"minutes": -1}, {"minutes": None}]

accepted = []
rejected = 0

for row in rows:
    value = row.get("minutes")

    if type(value) is int and 0 <= value <= 60:
        accepted.append(value)
    else:
        rejected = rejected + 1

print(accepted)
print(rejected)
```

## Follow the evidence

### Step or case · Evidence · Meaning

**Step or case:** Recorded zero

**Evidence:** 0

**Meaning:** Accept as a measurement

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**Step or case:** Absent or None

**Evidence:** Unknown

**Meaning:** Hold for review

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**Step or case:** Negative integer

**Evidence:** Outside range

**Meaning:** Reject under this schema

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Read each row from left to right and connect the evidence to the stated rule.

dict.get returns None when this key is absent. An explicit None also represents unknown in this dataset. append adds an item to the end of a list. The script prints [0, 12] and then 3. The exact type check prevents comparing None or text with the numeric range.

### Keep zero separate from unknown

A recorded 0 says the measured duration was zero. A missing field says we do not have the duration. They are different statements. Trace each record with three columns: raw value, decision and reason. Keep 0 and 12; reject the absent value, -1 and None. The accepted list therefore has two values and the rejection count is three. Do not turn an unknown value into a number just to make the calculation run.

### Connect the picture and the explanation

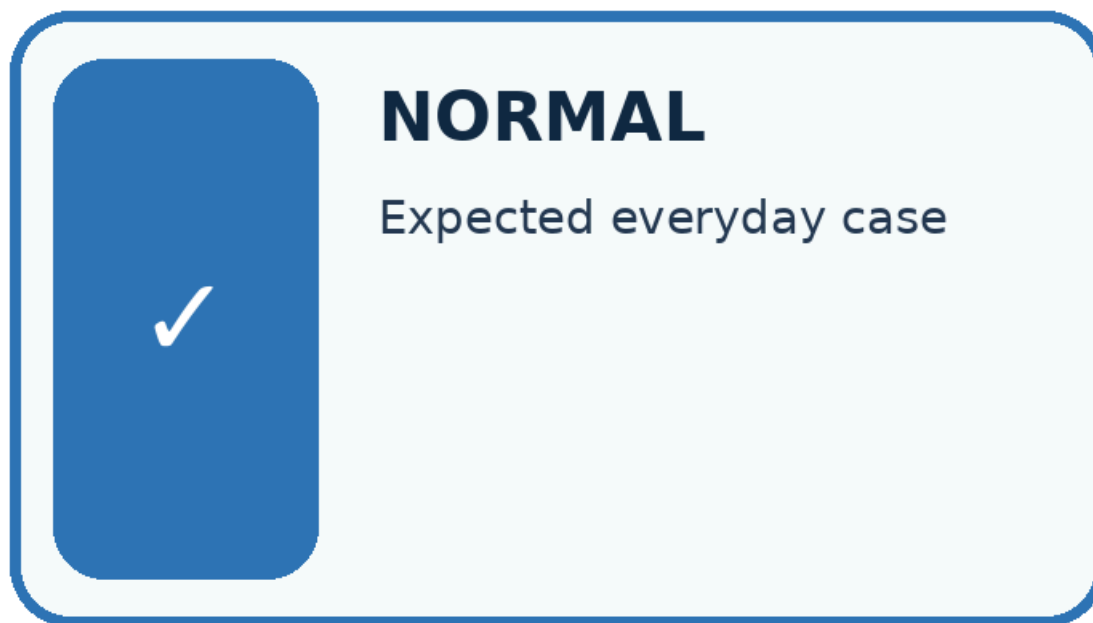
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

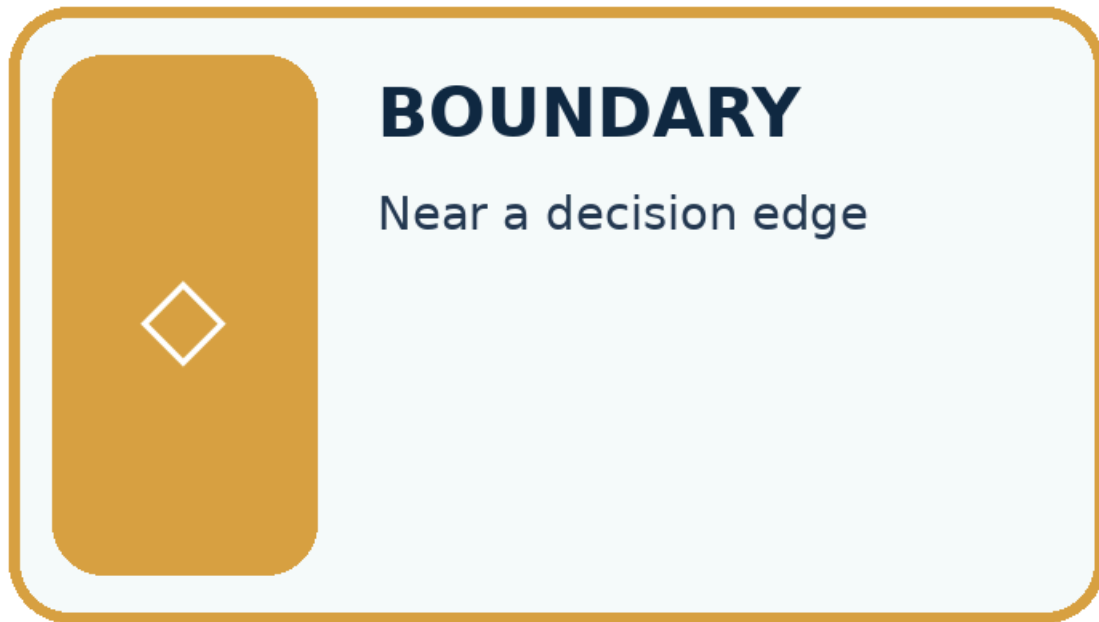
## 4. Compare the cases

### CASE GALLERY

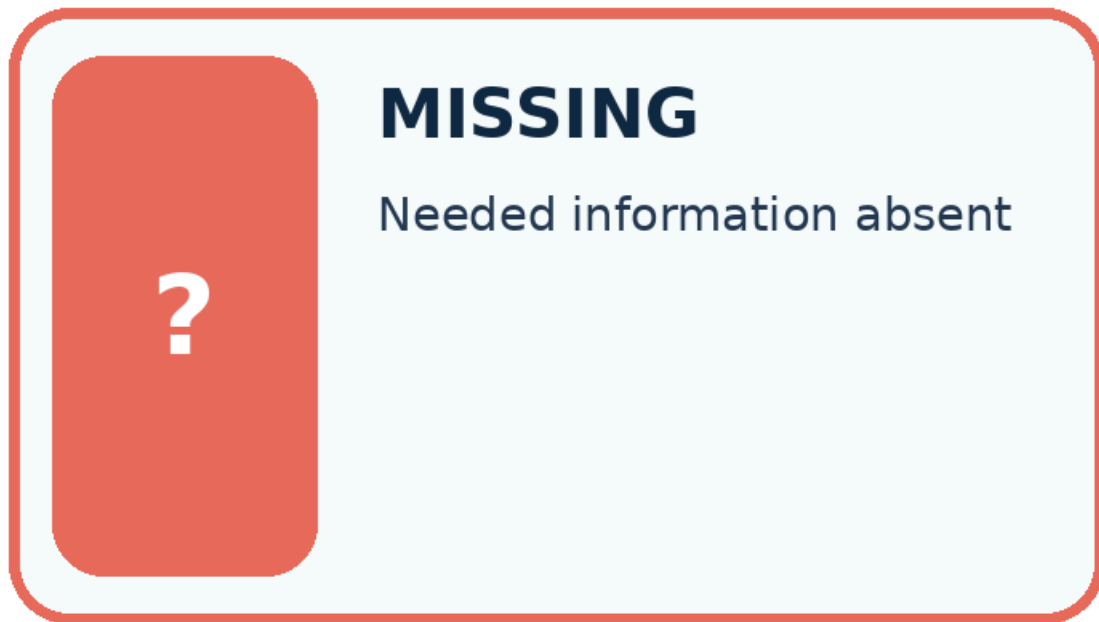
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.





MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

**Write or draw your reasoning and evidence.**

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### Change one thing in the pictured case

Change: Replace text "4" with integer 4 through an authorised conversion.

Prediction: Accepted values become [0,4,4]; three rows remain rejected.

Reason: The converted value now satisfies both type and range checks.

## 5. Words that unlock the idea

### VOCABULARY LAB

#### Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

**1**

## **MISSING DATA**

required value is unknown  
or absent

**2**

## **DEFAULT**

preselected replacement  
value

3

## ABSTAIN

decline to decide

4

## IMPUTE

estimate a missing value  
using a stated method

**Word · My own example · Not this / boundary**

**Word:** missing data

**My own example:** Information that was not recorded or is unavailable, which is different from an observed zero.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** default

**My own example:** The value or behaviour used when no alternative is specified.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** abstain

**My own example:** Decline to make a prediction when the available evidence does not meet the decision rule.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** impute

**My own example:** Replace a missing value using a stated method; an imputed value is an estimate, not an observation.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

## Words used in the closer explanation

### Word · Meaning

**Word:** Missing value

**Meaning:** A value that was not recorded

\_\_\_\_\_

**Word:** Validation

**Meaning:** Checking data against explicit rules

\_\_\_\_\_

**Word:** Rejection

**Meaning:** Holding a record out because it fails a rule

\_\_\_\_\_

**Word:** Coverage

**Meaning:** Which intended records or groups are represented

\_\_\_\_\_

## 6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 18 • WORKED EVIDENCE

- 1 Reader’s length preference is missing.
- 2 Ask: best when the user can answer.
- 3 Abstain: safest if the field is essential; default: convenient but may misrepresent.
- 4 Record which policy was used so the explanation stays truthful.

- 1. Reader’s length preference is missing.
- 2. Ask: best when the user can answer.
- 3. Abstain: safest if the field is essential; default: convenient but may misrepresent.
- 4. Record which policy was used so the explanation stays truthful.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Add {"minutes": 60} and {"minutes": "5"}. Predict the accepted list and rejection count.

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**Learn from the worked case**

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

**A pictorial worked case: Missing Data**

A fictional catalogue records counts of available copies under a strict input contract.

**The facts we will use**

Accept count only if type(value) is int and  $0 \leq \text{value} \leq 8$ . Missing fields are invalid, and Booleans are not counts.

Rows have count 0, count 4, absent count, count "4", count True and count 9.

## Missing Data

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Retrieve carefully

Use `row.get("count")`; absent field gives None, distinct from 0.



#### 2. Check exact type

Integers 0,4 pass; text "4", Boolean True and None fail.



#### 3. Check range

Integer 9 fails maximum 8.



#### 4. Report validation

Accepted counts [0,4]; four rows rejected with recorded reasons.

### CHANGE ONE THING

Replace text "4" with integer 4 through an authorised conversion.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Retrieve carefully

**What happens and why:** Use `row.get("count")`; absent field gives None, distinct from 0.

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**Step:** 2. Check exact type

**What happens and why:** Integers 0,4 pass; text "4", Boolean True and None fail.

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**Step: 3.** Check range

**What happens and why:** Integer 9 fails maximum 8.

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**Step: 4.** Report validation

**What happens and why:** Accepted counts [0,4]; four rows rejected with recorded reasons.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

## Why this result follows

Testing if value would wrongly discard zero, which is a real valid count. Python's `isinstance(True,int)` is true, so the stated exact-integer contract uses `type(value)` is int to reject Boolean input. Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero.

## What this example does not establish

Type and range checks cannot confirm whether the real shelf count was measured accurately.

# 7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



## COMPARE • DIAGNOSE

Weak idea: Blank values can safely be treated as zero without explanation.  
Strong move: locate evidence, change one cause, and retest.

**Evidence record · Expected · Actual · Likely cause / next test**

**Evidence record:** normal case

**Expected:** works

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** boundary case

**Expected:** stated rule

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** missing input

**Expected:** safe response

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** fresh transfer

**Expected:** generalizes

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

Repair one cause. Why is this repair better than changing several things?

**Write or draw your reasoning and evidence.**

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## A closer explanation

Rejection can change which sessions remain represented. Report 2 accepted of 5 supplied here. A summary of accepted rows does not automatically describe the missing or rejected sessions.

**Only 2 of 5 supplied durations were accepted. Why should the report include this?**

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## 8. Practise together

### LOGIC STUDIO

#### Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



### TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence\_record PREDICT expected\_result RUN one normal case and one boundary case  
COMPARE actual\_result with expected\_result IF different: locate first cause; change one factor; RETEST  
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

**Write or draw your reasoning and evidence.**

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## A closer explanation

Using if value would discard the legitimate zero and accept some invalid nonempty text. Test the written type and range rules explicitly. Filling an absent duration with 0 falsely claims an observed zero.

### Practise with less help: Trace filtering

Use workbook W2 for the connected example “Missing values and validation”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

#### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Missing values and validation. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

#### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

## EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



### PREDICT → TEST → RECORD → EXPLAIN

Run the same four incomplete profiles through ask, abstain, and default policies. Compare usefulness, risk, and user effort.

Run the same four incomplete profiles through ask, abstain, and default policies. Compare usefulness, risk, and user effort.

**Test / input · Prediction · Actual · Evidence / calculation · Change or keep?**

**Test / input:** normal

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** boundary

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** missing

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** fresh transfer

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

What did the hardest test teach you that the normal test did not?

**Write or draw your reasoning and evidence.**

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**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



## MASTERY LADDER

1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
  - 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
  - 3 • APPLY: Solve a different example independently.
  - 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
  - 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

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# 11. Connect your project

## CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



## TRANSPARENT BOOK RECOMMENDER

Use 'no preference' only when the reader confirms it; otherwise ask or explain uncertainty.

**THIS WEEK'S CHECKPOINT** Use 'no preference' only when the reader confirms it; otherwise ask or explain uncertainty.

**SYSTEM MAP** Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

**Write or draw your reasoning and evidence.**

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What should remain under human control?

**Write or draw your reasoning and evidence.**

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**RESPONSIBLE DESIGN** Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

**Evidence for your project**

Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing.

## 12. Show what you know

### INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



**INDEPENDENT PROOF**  
Compare ask, abstain, and default strategies.

CHALLENGE Compare ask, abstain, and default strategies.

1. My fresh example, trace, calculation, or design:

**Write or draw your reasoning and evidence.**

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2. My evidence that it works (include a boundary or missing-data case):

**Write or draw your reasoning and evidence.**

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3. One limitation, fairness concern, or situation needing human review:

**Write or draw your reasoning and evidence.**

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SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

## 13. Study a complete case

### Week 18 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

#### The situation

Genre is known, but page count is missing and length affects recommendation score.

#### The question

Choose transparent handling.

#### Worked explanation

Retrieve or ask for the count, or use a documented partial-score policy with uncertainty. Do not silently insert zero.

#### Explain it in your own words

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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## Trace the supplied case visually

### Missing is different from zero

#### Evidence or stage · Value or result

**Evidence or stage:** Retrieve

**Value or result:** Look up the page count

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**Evidence or stage:** Ask

**Value or result:** Request the missing information

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**Evidence or stage:** Partial score

**Value or result:** Use a documented policy with uncertainty

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Genre is known, but page count is missing and length affects recommendation score.

Choose transparent handling.

Retrieve or ask for the count, or use a documented partial-score policy with uncertainty. Do not silently insert zero.

## 14. Try a new case independently

### Independent case: Validate independently

Write a filter for count values that must be integers from 0 to 10. Process {count: 3}, {}, {count: 0}, {count: 11}, {count: "3"}, using quoted keys in Python. Report accepted values and rejected count.

#### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## Week 18 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A book's page count is blank. Is blank the same as zero? Give an honest way for a short-book recommender to respond.

My answer, with a trace, calculation or supporting evidence

**Write or draw your reasoning and evidence.**

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## Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

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## Connect to your project

Use 'no preference' only when the reader confirms it; otherwise ask or explain uncertainty.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

### Transfer practice from the original programme

Scores award 2 for preferred genre and 1 for short length. A book matches genre but page count is missing. Compare rejecting it, assigning zero length points and asking for pages.

**Explain the mechanism, evidence and one limit.**

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### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.